Fire-aware robot dogs

Machine-learning wildfire forecasts for safe navigation and scouting

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Why

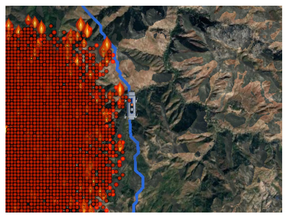
The edge of a wildfire is the most dangerous place for a firefighter.

Robot dogs (four-legged robots) can climb steep, rocky, smoky ground where trucks get stuck and drones can't see.

But a dog that steers by where the fire is now can walk into where it will be.

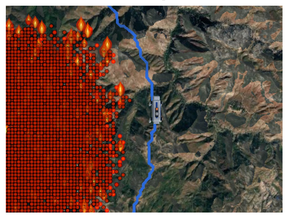
Plan on the predicted fire

Planned on the fire now



In the flames
An hour in, the fire has reached the route.

Planned on the prediction



Still clear
Same moment: this route keeps 100 m from the predicted fire.

The dog stays at least 100 m from where the fire is predicted to be, with 30 min in hand.

It scouts from the closest spot that keeps that distance and still has a way home. When its sensors show the real fire differs, the prediction reruns and the dog re-plans.

Simulator screenshots, Big Sur: same fire, same start and target, one hour after the dog sets out.

How we predict the fire

Fire history + conditions
246,000 NASA satellite fire detections from 30 wildfires; weather, terrain, land cover

→

Machine learning + fire physics
XGBoost adjusts a Rothermel spread model

→

When each spot burns
Every 50 m spot, in 15 min steps

Small and fast, so the dog can re-plan on the spot.

31 KB
The whole model. Smaller than one photo on this poster.

0.1 s
To predict all 360,000 spots on the map (desktop computer).

Anywhere
Small edge computers like a Raspberry Pi. No internet or GPU.

Close enough to help, safe enough to return

430 simulated missions per dog

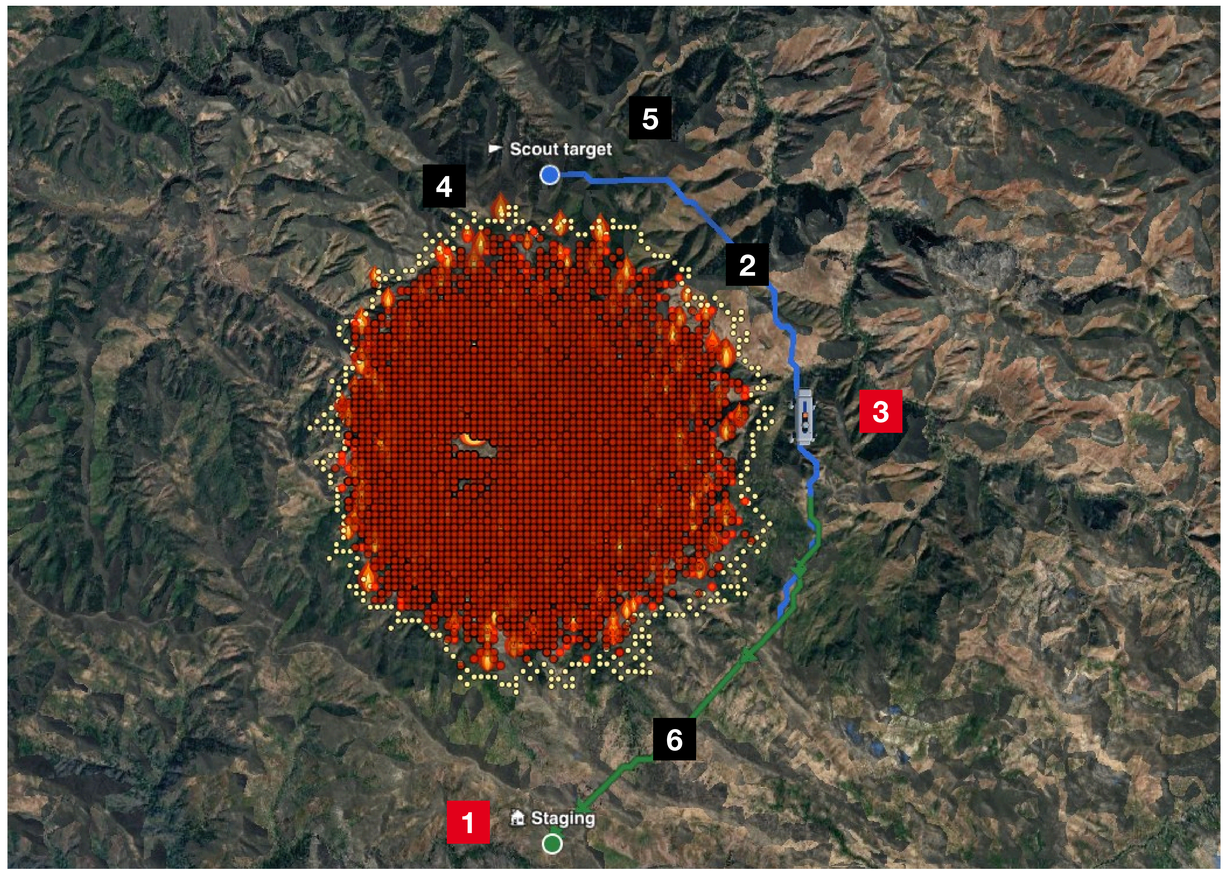
	Reached a spot to scout	Caught by the fire
Reacts, keeps 500 m away	0%	0%
Reacts, keeps 200 m away	36%	0.9%
Predicts, no margin	100%	44%
Predicts, 100 m + 30 min + sensing	95%	0%

Reacting gets close only when it's lucky. Predicting knows when it's safe to go.

Each mission: walk from staging toward a target at the fire's edge, scout, come home. The predicting dog scouts from the closest spot that keeps 100 m from the predicted fire. Simulated on made-up terrain; the real fire runs hotter and drier than forecast, sometimes with a wind shift. No physical robot.

One afternoon at a Big Sur fire

A simulated mission on real terrain. Follow the dog from stop 1 to stop 6. How it decides at each stop: right panel.



1 **2:45 PM** Launch
Steep, brushy slopes, burning since midnight. Too dangerous for a crew, so the dog leaves staging.

2 **2:45 PM** Plan
The **blue** route keeps 100 m from the forecast fire, with 30 min in hand, slower in brush and uphill.

3 **3:15 PM** Wind shift
The real fire turns; the forecast doesn't know, so the prediction needs a rerun. The **green** escape path stays open.

4 **4:30 PM** Scout
It reaches its scout spot, as close as the 100 m rule allows.

5 **4:30 PM** Measure
Its sensors find wind from the northwest, not the west, and 40 °C, not 35. The prediction reruns: the fire is coming, so the dog re-plans and heads home another way.

6 **~6:45 PM** Home safe
Back before the fire arrives. The crew plans with a better forecast.

Know where the fire will be, not just where it is.

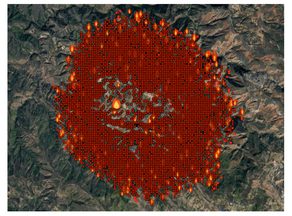
Safer crews
Fewer firefighters sent into danger to look.

Robot stays safe
Routes and exits keep it from getting trapped.

Smarter sensing
Measure where it changes decisions.

How the dog decides

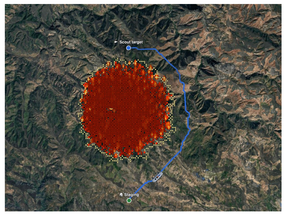
1 Predict



50 m cells **15 min** steps

Forecasts when each cell will burn.

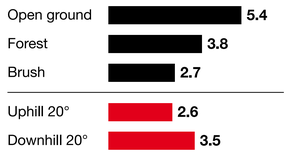
2 Safe route



Blue route to the scout target

Keeps 100 m from the forecast fire, with 30 min in hand.

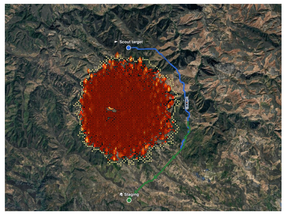
2b Route speed



Walking speed in **km/h**

The route is timed at these speeds. The dog never crosses water or slopes over 35°.

3 Escape path



Green escape path home

Aims for a way home that stays 100 m clear for an hour.

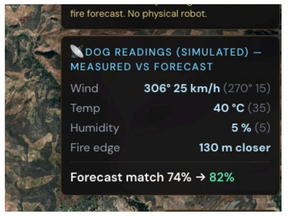
4 Scout



Wind, temperature, humidity, fire edge

Measures what the forecast only guessed.

5 Feedback



Launch **83%** Shift **74%** Report **82%**

The readings improve the next forecast.

Simulator views on real terrain. Yellow dotted edge: the forecast the dog plans on; the real fire is hidden from it until its sensors see it. Robot image: design concept; sensors are simulated.

Status and next

Done	Next
ML fire forecast trained and tested	Forecast + planner on the dog's controller
3D browser simulator on real terrain	Real sensors feeding the forecast
Containment planner (search)	Scout the planned containment line
Robot-dog loop in mock simulation	Tests on real robot dogs

Open-source code and demo video

Data and references

Code: github.com/sxz2012/wildfire-simulation

NASA FIRMS (VIIRS active-fire detections); Open-Meteo weather archive; ESA WorldCover 10 m land cover; Cesium World Terrain; Google Photorealistic 3D Tiles

Rothermel, R. C. (1972). *A mathematical model for predicting fire spread in wildland fuels.* USDA Forest Service INT-115.

Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD '16.*